

AUCA — Introduction to Big Data Analytics

Week 1 Lab Report — Python Foundations

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Introduction

This week's lab introduced the basic Python foundations needed for data analysis. I created variables using different data types, including strings, integers, floats, and Boolean values. I also practiced type conversion by converting a number stored as text into an integer.

What I Built

During the lab, I worked with strings, lists, and dictionaries. I practiced using f-strings, string methods, list operations, and dictionary operations. I also analyzed a dataset containing 20 AUCA students. The analysis included calculating the number of students, class average, highest and lowest scores, score range, individual student records, and the number of students from selected districts.

The class dataset had an average score of **76.7**. The highest score was **94**, achieved by **Sandrine Umutoni**, while the lowest score was **55**, achieved by **David Hakizimana**.

What I Learned

One thing that surprised me was that `"3" + "4"` produces `"34"` instead of `7`. This happened because the values are strings, so Python joins them together rather than performing numerical addition. This helped me understand why identifying and converting data types is important when working with real-world datasets.

I also learned that a dataset containing only 20 students is not considered Big Data because it is too small in volume and does not demonstrate the scale and complexity normally associated with Big Data. It mainly contains structured information and does not clearly demonstrate significant Volume, Velocity, or Variety.

Conclusion

Overall, the lab helped me understand the basic Python concepts required for working with data. I practiced storing, modifying, and analyzing data and gained a better understanding of how Python can be used as a foundation for Big Data Analytics.